# INVERBERVIE, United Kingdom

WMO: **030880**

Lat: **56.8519N** Lon: **2.2658W** Elev: **134** StdP: **99.73** Time Zone: **0.00 (EUW)** Period: **94-19** WBAN: **99999**

## Annual Heating, Humidification, and Ventilation Design Conditions

| | Coldest Month | Heating DB | | Humidification DP/MCDB and HR | | | | | | Coldest Month WS/MCDB | | | | MCWS/PCWD to 99.6% DB | | WSF | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 99.6% | | | 99% | | | 0.4% | | 1% | | | | | |
| | | 99.6% | 99% | DP | HR | MCDB | DP | HR | MCDB | WS | MCDB | WS | MCDB | MCWS | PCWD | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (1) | **1** | **-2.4** | **-1.2** | **-5.2** | **2.5** | **0.3** | **-4.2** | **2.7** | **0.8** | **22.2** | **6.2** | **20.2** | **5.6** | **6.2** | **310** | **0.811** | (1) |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| | Hottest Month | Hottest Month DB Range | Cooling DB/MCWB | | | | | | Evaporation WB/MCDB | | | | | | MCWS/PCWD to 0.4% DB | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 0.4% | | 1% | | 2% | | 0.4% | | 1% | | 2% | | | | |
| | | | DB | MCWB | DB | MCWB | DB | MCWB | WB | MCDB | WB | MCDB | WB | MCDB | MCWS | PCWD | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (2) | **8** | **5.5** | **19.2** | **15.4** | **17.9** | **14.7** | **16.8** | **14.1** | **16.6** | **18.2** | **15.7** | **17.0** | **14.9** | **16.1** | **6.8** | **250** | (2) |

| | Dehumidification DP/MCDB and HR | | | | | | | | | Enthalpy/MCDB | | | | | | Extreme Max WB | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0.4% | | | 1% | | | 2% | | | 0.4% | | 1% | | 2% | | | |
| | DP | HR | MCDB | DP | HR | MCDB | DP | HR | MCDB | Enth | MCDB | Enth | MCDB | Enth | MCDB | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (3) | **15.9** | **11.4** | **17.2** | **15.1** | **10.9** | **16.3** | **14.3** | **10.3** | **15.5** | **46.9** | **18.4** | **44.3** | **17.0** | **42.2** | **16.2** | **20.9** | (3) |

## Extreme Annual Design Conditions

| | Extreme Annual WS | | | | Extreme Annual Temperature | | | | n-Year Return Period Values of Extreme Temperature | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Mean | | Standard Deviation | | n=5 years | | n=10 years | | n=20 years | | n=50 years | | |
| | 1% | 2.5% | 5% | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| | (a) | (b) | (c) | | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (4) | **17.7** | **15.2** | **13.5** | DB | **-3.8** | **23.2** | **0.9** | **1.3** | **-4.4** | **24.2** | **-5.0** | **25.0** | **-5.5** | **25.7** | **-6.2** | **26.7** | (4) |
| (5) | | | | WB | **-4.4** | **18.8** | **1.1** | **1.1** | **-5.2** | **19.6** | **-5.9** | **20.3** | **-6.5** | **20.9** | **-7.3** | **21.8** | (5) |

## Monthly Climatic Design Conditions

| | | | | Annual | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (6) | Temperatures, Degree-Days and Degree-Hours | | DBAvg | **8.2** | **3.7** | **3.8** | **5.0** | **6.6** | **8.7** | **11.4** | **13.5** | **13.6** | **11.9** | **9.3** | **6.1** | **4.2** | (6) |
| (7) | | | DBStd | **4.30** | **2.24** | **2.54** | **2.67** | **2.35** | **2.43** | **2.10** | **2.00** | **1.76** | **1.95** | **2.38** | **2.66** | **2.74** | (7) |
| (8) | | | HDD10.0 | **1036** | **195** | **173** | **157** | **104** | **54** | **9** | **1** | **1** | **4** | **42** | **118** | **178** | (8) |
| (9) | | | HDD18.3 | **3701** | **453** | **406** | **414** | **352** | **298** | **207** | **149** | **145** | **192** | **282** | **367** | **437** | (9) |
| (10) | | | CDD10.0 | **376** | **0** | **0** | **1** | **3** | **14** | **52** | **110** | **113** | **63** | **18** | **2** | **0** | (10) |
| (11) | | | CDD18.3 | **1** | **0** | **0** | **0** | **0** | **0** | **0** | **1** | **0** | **0** | **0** | **0** | **0** | (11) |
| (12) | | | CDH23.3 | **1** | **0** | **0** | **0** | **0** | **0** | **0** | **1** | **0** | **0** | **0** | **0** | **0** | (12) |
| (13) | | | CDH26.7 | **0** | **0** | **0** | **0** | **0** | **0** | **0** | **0** | **0** | **0** | **0** | **0** | **0** | (13) |
| (14) | Wind | | WSAvg | **6.5** | **8.1** | **7.8** | **7.1** | **6.1** | **5.7** | **5.4** | **4.7** | **5.1** | **6.1** | **7.2** | **7.3** | **7.8** | (14) |
| (15) | Precipitation | | PrecAvg | **836** | **83** | **57** | **55** | **56** | **58** | **57** | **71** | **68** | **66** | **94** | **87** | **83** | (15) |
| (16) | | | PrecMax | **1034** | **221** | **119** | **149** | **145** | **126** | **136** | **141** | **146** | **240** | **224** | **195** | **201** | (16) |
| (17) | | | PrecMin | **522** | **15** | **6** | **15** | **0** | **10** | **15** | **20** | **11** | **15** | **7** | **25** | **19** | (17) |
| (18) | | | PrecStd | **122** | **47** | **37** | **28** | **40** | **27** | **34** | **38** | **34** | **46** | **51** | **49** | **44** | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% | DB | | **11.0** | **12.1** | **14.2** | **15.4** | **17.4** | **19.6** | **21.9** | **20.7** | **19.9** | **15.8** | **12.9** | **11.7** | (19) |
| (20) | | | MCWB | | **9.2** | **10.0** | **10.1** | **11.2** | **13.4** | **14.8** | **17.2** | **16.6** | **16.1** | **13.0** | **11.3** | **10.3** | (20) |
| (21) | | 2% | DB | | **9.4** | **10.5** | **11.8** | **12.9** | **15.4** | **17.4** | **19.3** | **18.7** | **17.2** | **14.2** | **11.5** | **10.2** | (21) |
| (22) | | | MCWB | | **8.1** | **8.7** | **8.6** | **9.9** | **12.3** | **13.6** | **15.9** | **15.1** | **14.0** | **12.2** | **10.4** | **9.2** | (22) |
| (23) | | 5% | DB | | **8.1** | **8.8** | **10.1** | **11.4** | **13.6** | **16.0** | **17.9** | **17.5** | **15.8** | **13.1** | **10.7** | **9.1** | (23) |
| (24) | | | MCWB | | **7.0** | **7.2** | **7.6** | **8.7** | **10.9** | **13.0** | **15.0** | **14.7** | **13.4** | **11.7** | **9.8** | **8.3** | (24) |
| (25) | | 10% | DB | | **7.1** | **7.5** | **8.7** | **10.1** | **12.3** | **14.6** | **16.6** | **16.4** | **14.8** | **12.4** | **9.9** | **8.2** | (25) |
| (26) | | | MCWB | | **6.3** | **6.0** | **6.8** | **8.1** | **10.1** | **12.4** | **14.5** | **14.2** | **12.9** | **11.2** | **9.2** | **7.5** | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% | WB | | **9.7** | **10.4** | **10.7** | **12.1** | **14.3** | **16.3** | **18.3** | **17.5** | **16.5** | **13.9** | **11.7** | **10.7** | (27) |
| (28) | | | MCDB | | **10.6** | **11.9** | **13.3** | **14.4** | **16.4** | **18.2** | **20.4** | **19.4** | **19.0** | **15.1** | **12.2** | **11.3** | (28) |
| (29) | | 2% | WB | | **8.4** | **8.8** | **9.2** | **10.5** | **12.7** | **14.7** | **16.8** | **16.2** | **15.0** | **12.9** | **10.8** | **9.5** | (29) |
| (30) | | | MCDB | | **9.2** | **10.3** | **11.1** | **12.3** | **14.6** | **16.4** | **18.5** | **17.6** | **16.4** | **13.6** | **11.3** | **10.1** | (30) |
| (31) | | 5% | WB | | **7.2** | **7.5** | **8.0** | **9.2** | **11.6** | **13.7** | **15.8** | **15.5** | **14.1** | **12.2** | **10.1** | **8.5** | (31) |
| (32) | | | MCDB | | **7.9** | **8.6** | **9.7** | **10.8** | **13.1** | **15.2** | **17.1** | **16.7** | **15.2** | **12.8** | **10.5** | **9.1** | (32) |
| (33) | | 10% | WB | | **6.4** | **6.4** | **7.0** | **8.3** | **10.6** | **12.8** | **15.0** | **14.8** | **13.4** | **11.6** | **9.3** | **7.6** | (33) |
| (34) | | | MCDB | | **7.0** | **7.3** | **8.5** | **9.6** | **11.9** | **14.1** | **16.2** | **15.8** | **14.2** | **12.2** | **9.9** | **8.2** | (34) |
| (35) | Mean Daily Temperature Range | | MDBR | | **3.8** | **4.4** | **5.2** | **5.3** | **5.6** | **5.5** | **5.5** | **5.5** | **5.3** | **4.7** | **4.3** | **3.9** | (35) |
| (36) | | 5% DB | MCDBR | | **5.4** | **5.7** | **7.5** | **7.5** | **7.5** | **7.3** | **7.4** | **7.1** | **6.8** | **5.4** | **4.7** | **4.9** | (36) |
| (37) | | | MCWBR | | **5.2** | **4.7** | **5.4** | **5.2** | **5.1** | **4.9** | **4.7** | **4.4** | **4.7** | **4.5** | **4.6** | **4.7** | (37) |
| (38) | | 5% WB | MCDBR | | **5.0** | **5.5** | **7.0** | **6.8** | **6.8** | **6.4** | **6.5** | **5.8** | **5.8** | **4.9** | **4.4** | **4.8** | (38) |
| (39) | | | MCWBR | | **4.9** | **4.7** | **5.3** | **5.1** | **5.1** | **4.8** | **4.7** | **4.2** | **4.5** | **4.5** | **4.5** | **4.9** | (39) |
| (40) | Clear-Sky Solar Irradiance | taub | | | **0.304** | **0.325** | **0.353** | **0.374** | **0.370** | **0.371** | **0.373** | **0.362** | **0.347** | **0.334** | **0.305** | **0.286** | (40) |
| (41) | | taud | | | **2.289** | **2.387** | **2.359** | **2.329** | **2.393** | **2.414** | **2.411** | **2.449** | **2.498** | **2.463** | **2.341** | **2.230** | (41) |
| (42) | | Ebn at Noon | | | **577** | **733** | **803** | **840** | **868** | **870** | **860** | **846** | **808** | **715** | **573** | **485** | (42) |
| (43) | | Edh at Noon | | | **51** | **69** | **91** | **108** | **108** | **107** | **106** | **95** | **79** | **64** | **47** | **40** | (43) |
| (44) | All-Sky Solar Radiation | RadAvg | | | **0.46** | **1.11** | **2.18** | **3.62** | **4.87** | **4.80** | **4.63** | **3.70** | **2.56** | **1.35** | **0.60** | **0.33** | (44) |
| (45) | | RadStd | | | **0.06** | **0.10** | **0.26** | **0.35** | **0.27** | **0.38** | **0.36** | **0.25** | **0.19** | **0.13** | **0.06** | **0.04** | (45) |

## Historical Trends

| | | DBAvg | Heating | | Cooling | | | Degree-Days | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 99% DB | 99% DP | 1% DB | 1% WB | 1% DP | HDD10.0 | HDD18.3 | CDD10.0 | CDD18.3 | |
| | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | |
| (46) | Station Only | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (46) |
| (47) | Regional (32 neighbors) | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (47) |

Nomenclature: See separate page